

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

Nov 2013

EE304 Industrial Instrumentation

Date: 20.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define the following: [07]
a) Accuracy b) Precision c) Repeatability d) Reproducibility e) Speed of Response
f) Fidelity g) Dynamic Lag

Q - 2 (a) Describe in detail principle, construction, working, advantages, disadvantages and range of Resistance Temperature Detector (RTD) [07]

(b) Describe in detail principle, construction, working, advantages, disadvantages and range of capacitance level measurement [07]

OR

Q - 2 (a) Describe in detail principle, construction, working, advantages, disadvantages and range of Air purge/ bubbler method [07]

(b) Describe in detail principle, construction, working, advantages and disadvantages of turbine flow meter [07]

Q - 3 (a) Define error. Describe in detail classification of error. Also state various sources of error. [07]

(b) Classify and explain the types of strain gauge in detail. [07]

OR

Q - 3 (a) List out various techniques for measuring speed and explain principle, construction, working, advantages and disadvantages of one of the techniques [07]

(b) List various methods used for angular displacement measurement and explain principle, construction, working, advantages and disadvantages of any one in detail [07]

SECTION – II

Q - 4 (a) Define the following: [07]
a) Temperature b) Absolute Pressure c) Gauge Pressure d) Atmospheric Pressure
e) Sensitivity f) Drift g) Dead Zone

Q - 5 (a) Explain Resistive hygrometers [07]

(b) Explain X-Y recorders [07]

OR

Q - 5 (a) Describe in detail principle, construction, working, advantages and disadvantages of electromagnetic flow meter [07]

(b) Describe in detail principle, construction, working, advantages and disadvantages of radiation pyrometer [07]

Q - 6 (a) Attempt any two: [14]

- 1) Hot wire anemometer
- 2) Ultrasonic flow meter
- 3) Strip chart recorder
- 4) Optical pyrometer

OR

Q - 6 (a) Discuss: types and characteristics of thermocouples [14]